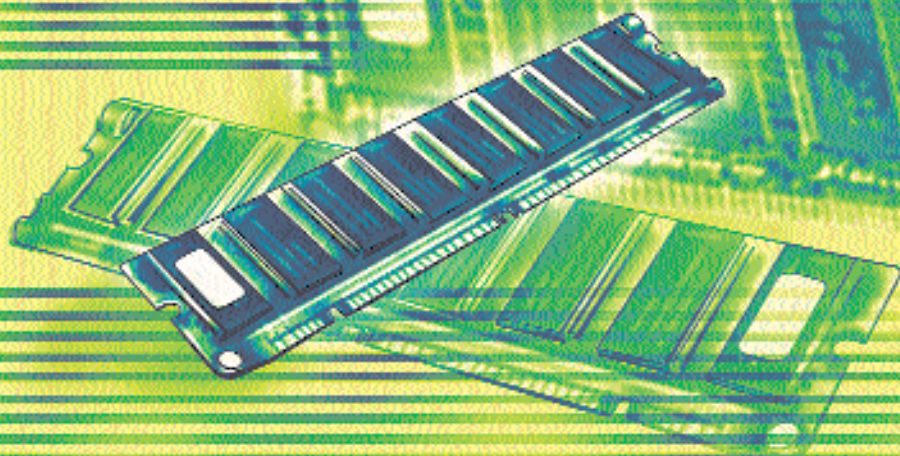




RAM

This era belongs to DDR RAM. DDR started with a speed of 266 MHz, and now has clocked speeds of 1066 MHz (DDR2). The market has recently witnessed changes such as the implementation of dual-channel RAM.

Questions to ask the dealer

What type of packaging does the module come in?

Prefer RAM modules that come in tamper-proof packages. Also check with the dealer about the hologram, if any, to verify the originality of the RAM module.

Does the RAM come with a warranty? If so, how long is the warranty period?

The warranty is manufacturer-dependent. Some manufacturers give a three-year warranty, and others give only one year. It's therefore important to ask the dealer about the warranty.

Interesting facts

Static RAM maintains its data for as long as the memory chip is supplied with power. Dynamic RAM needs continuous re-writing of data so as to maintain its

content. Static RAM is faster than Dynamic RAM for this reason, but is very expensive.

DRAM is used for fabricating system memory as it is cheap. The RAM we use today is an evolution of DRAM. The earliest of this family was FPM DRAM (Fast Page Mode DRAM); it was designed to enhance system performance when reading or writing bursts of data. Later came EDO (Extended Data Out) RAM, which had a feature by which it could start a new access cycle while maintaining the data output of the previous cycle.

The demand for more speed and performance led to the development of SDRAM (Synchronous DRAM). DRAM is asynchronous; SDRAM synchronises with the clock pulses when responding to the incoming commands. This changed the method of functioning of DRAM by enabling more complex patterns of operation. But the pipelining involved here led to a slight delay termed latency, which means that data is accepted even before it is actually written to the address. Similarly, requested data is read a few clock pulses after issuing the read command.

SDRAM, like most logic circuits, responds only to the low-to-high transition of a clock pulse. This led to the evolution of DDR (Double Data Rate) RAM, which reacts to both the low-to-high and high-to-low transitions of the clock pulse—so it can read/write data twice in a single clock pulse.



Usage Tips

■ In the BIOS, select the highest clock setting supported by the system. Prefer a low-latency memory module if you intend to overclock it for a better gaming experience.

■ A lower CAS latency setting leads to a slight improvement in performance.

■ While inserting a RAM module, match the notch on the RAM chip with that on the motherboard RAM slot. Hold the RAM at the ends and use your thumb to push it into the slot. You will hear the lock on the motherboard slot snap, which indicates that the RAM is fixed. If you are not able to push it in, don't force the module—instead, cross-check the position of the notches and also make sure you have opened the locks of the slot.

■ When cleaning a RAM module, do not use cloth—especially artificial fabric such as nylon—because it causes static charges that may damage the module. Use an eraser to clean the system bus interfacing pins. ■

Points To Ponder

What type of RAM is supported by my system?

You need to first figure out what basic type of RAM (SDRAM, DDR RAM or RDRAM) is supported by your system. This information is available in your motherboard's manual. Second, find out the speed of the RAM supported by your motherboard. The speed may be 266, 333 and 400 MHz for DDR, and 533 or 800 MHz for DDR2. For RDRAM the speeds are 533 and 800 MHz, and the older SDRAM's clock ticks at 66, 100, 133 and 166 MHz.

How much memory do I need?

Our answer is "as much as you can afford." If that doesn't make sense, look at your needs. If you use your system mainly for Internet brows-

ing and such, you shouldn't need more than 256 MB. Graphics professionals should look for at least 512 MB of RAM. 512 MB will also result in a slight performance increase for those with moderate computing requirements. Gamers, of course, should start with nothing less than 1 GB.

What should I consider when upgrading RAM?

It is very important that the new RAM module match the older module(s). That is, if the clock speed of the new RAM module is different from that of the older one(s), the mismatch will result in frequent reboots. The POST screen displays the clock speed of the RAM below the memory count.